

Sarthak Kala

+1 619-953-8214 | skala@ucsd.edu | [LinkedIn](#) | [Github](#) | [Website](#)

EDUCATION

University of California, San Diego

Sep 2024 – Present (*Expected Dec 2025*)

Master of Science, Data Science

La Jolla, CA

- GPA: 3.9/4.0
- Coursework: Safety in Generative AI, Deep Generative Models, Machine Learning, Recommender Systems, ML Systems, Advanced Data Mining

INTERNSHIP

Juniper Networks

Jun 2025 - Sep 2025

Software Engineering Intern - AI/ML

Sunnyvale, CA

- **AI Agents for Switch Testing:** Designed **AI agents** that converted switch documentation in natural language to different attack hypotheses, converted it into code and monitored switch performance.

EXPERIENCE

American Express Company

Feb 2022 - Aug 2024

Senior Data Scientist / Machine Learning Engineer

Gurgaon, India

- **LLM fine-tuning** and optimization for **retrieval-augmented generation** based Q&A systems, enabling marketing teams to efficiently query marketing plans, promotions and executive summaries. Implemented **quantization and pruning**, reducing computational costs by **35%**.
- Worked with AmEx AI Labs to build a **customer complaints segmentation** model using **Latent Dirichlet Allocation** (LDA) and **BERT** embeddings, reducing complaints response time by **75%**.
- Designed **Marketing Mix Models (MMM)** to analyze **\$60M+** in media ad spend, influencing C-level decisions on ad strategy. Led to **2x** increase in media budgets, and investment reduction in inefficient channels.
- Modernized MMM using **Bayesian inference** with **Markov Chain Monte Carlo (MCMC)** sampling to incorporate prior beliefs and quantify uncertainty in the model. Reduced manual intervention and model runtime by **60%**.
- Developed models to **quantify ad saturation**, enabling marketing teams to simulate budget allocation scenarios and optimize spend.
- Selected for a **high-potential leadership program**, leading a team of two analysts during my manager's maternity leave.

American Express Company

Jun 2019 - Jan 2022

Data Scientist

Gurgaon, India

- Generated **actionable insights** for referral digital marketing channels for a **\$4B+** annual business, across product, ad placement, financial, and demographic data to refine marketing strategies and boost ROI.
- Built a predictive model using **xGBoost** to identify high-potential referrers from existing customers, enabling better-targeted marketing, reducing ad spend, and driving **\$10M+** in incremental revenue.
- Designed and executed **A/B test** for referral marketing campaigns, resulting in **\$15M+** uplift in conversion rates across UK, France, Germany, Canada, and Australia markets.
- Built a **customer segmentation** model using **K-Means clustering**, leveraging clickstream and behavioral data to optimize retargeting campaigns and improve conversion rates by **12%**.
- Built scalable **ETL** pipelines using **Apache Spark**, ensuring efficient processing of referral and clickstream data.
- Designed and implemented containerized solutions using Docker, orchestrated microservices architectures with Kubernetes, and integrated CI/CD pipelines to streamline deployments, ensuring enhanced scalability, maintainability, and operational efficiency.

TECHNICAL SKILLS

Programming and Tools: Python, Apache Spark, SQL, Shell Scripting, C/C++, Kafka, Docker, Git, Kubernetes, AWS, Tableau, Hadoop, BigQuery

ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, PyMC, Matplotlib, Seaborn

Algorithms and Models: Regression, SVM, Clustering, Random Forests, XGBoost, Neural Networks, CNN, RNN, Bayesian methods, Attention, Hugging Face Transformers, Large Language Models (LLMs), LangChain, Llama, Retrieval-Augmented Generation (RAGs), LoRA, Quantization

RESEARCH PROJECTS

Causal Copilot Agent

Apr 2025 - Jun 2025

Research Assistant — Prof. Biwei Huang, Halicioglu Data Science Institute

UC San Diego

- Designed a pipeline of **AI agents** with **CrewAI** that autonomously generates, ranks, and validates causal hypotheses.

Intelligence prediction from fMRI

Jan 2025 - Mar 2025

Graduate Student Researcher — Prof. Armin Schwartzman, Halicioglu Data Science Institute

UC San Diego

- Used **Graph Convolutional Network** and **Graph Attention Network** based models on cortical surface region of the brain, predicting crystallized intelligence from adolescent brain fMRI imaging dataset.